

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026
Chemistry — SSC-I (9th Grade)
Syllabus: Chapter 8 — Energetics

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

General Instructions:

- Write your name, roll number and section clearly on the answer sheet.
- Section A is compulsory. Encircle the correct option on the question paper itself.
- In Section B, attempt **all 11** questions. For each question, attempt either the main question **OR** its alternative.
- In Section C, attempt **all 4** questions. For each question, attempt either the main question **OR** its alternative.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	A chemical reaction that proceeds with the <i>evolution</i> of heat is called:	Endothermic reaction	Exothermic reaction	Thermochemical reaction	Catalytic reaction	Ⓐ Ⓑ Ⓒ Ⓓ
2	If the ΔH value of a reaction is <i>negative</i> , the reaction is:	Endothermic	Exothermic	Neutral	Catalytic	Ⓐ Ⓑ Ⓒ Ⓓ
3	The enthalpy change for $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ is $\Delta H^\circ = -393.5$ kJ. This reaction is:	Endothermic	Exothermic	Reversible	Neutralization	Ⓐ Ⓑ Ⓒ Ⓓ
4	Bond <i>breaking</i> is always:	Exothermic	Endothermic	Catalytic	Thermochemical	Ⓐ Ⓑ Ⓒ Ⓓ
5	The enthalpy change of a reaction equals:	Sum of BDE of reactants only	Sum of BFE of products only	Sum of BDE of reactants minus sum of BFE of products	Sum of BFE of products minus sum of BDE of reactants	Ⓐ Ⓑ Ⓒ Ⓓ
6	Activation energy of a reaction must be _____ the average kinetic energy of reacting molecules for an effective collision:	lower than	equal to	greater than	unrelated to	Ⓐ Ⓑ Ⓒ Ⓓ
7	A catalyst increases the rate of a reaction by:	increasing activation energy	increasing enthalpy of reaction	lowering activation energy	decreasing enthalpy of reaction	Ⓐ Ⓑ Ⓒ Ⓓ
8	A balanced chemical equation that also shows the heat change of a reaction is called:	Ionic equation	Thermochemical equation	Half equation	Nuclear equation	Ⓐ Ⓑ Ⓒ Ⓓ
9	Which product is <i>not</i> released in aerobic respiration?	Carbon dioxide	Water	Lactic acid	Energy	Ⓐ Ⓑ Ⓒ Ⓓ
10	Which of the following reactions has a <i>positive</i> enthalpy change?	$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}, \Delta H^\circ = -393.5$ kJ	$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(g)}, \Delta H^\circ = -571.6$ kJ	$N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}, \Delta H^\circ = +180.5$ kJ	$C_{(s)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)}, \Delta H^\circ = -110.5$ kJ	Ⓐ Ⓑ Ⓒ Ⓓ
11	Lipids act as reserve energy stores because:	they dissolve in water	they can store very large amounts of energy	they increase activation energy	they are catalysts in the body	Ⓐ Ⓑ Ⓒ Ⓓ
12	In living organisms, the catalysts that catalyze chemical reactions are called:	Vitamins	Hormones	Enzymes	Lipids	Ⓐ Ⓑ Ⓒ Ⓓ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	Define chemical energetics. What is a thermochemical reaction? Give one example with its thermochemical equation.	3	OR	Define a chemical system and its surroundings. Give one example from the textbook to illustrate the difference between system and surroundings.	3
2(ii)	Define exothermic reaction. Write two examples of exothermic reactions with their thermochemical equations and ΔH values.	3	OR	Define endothermic reaction. Write two examples of endothermic reactions with their thermochemical equations and ΔH values.	3
2(iii)	State the sign convention for enthalpy change (ΔH). Why is ΔH negative for an exothermic reaction and positive for an endothermic reaction?	3	OR	What is standard enthalpy change? At what temperature and pressure is it measured? Write its symbol.	3
2(iv)	Classify the following processes as exothermic or endothermic: (a) freezing of water (b) combustion of methane (c) decomposition of limestone.	3	OR	Classify the following processes as exothermic or endothermic: (a) sublimation of dry ice (b) $H_2O_{(g)} \rightarrow H_2O_{(l)}$ (c) burning of fuels.	3

2(v)	Define bond dissociation energy and bond formation energy. State the relationship between them and the enthalpy change of a reaction.	3	OR	Explain why bond breaking always absorbs energy and bond forming always releases energy. How does this determine whether a reaction is exothermic or endothermic?	3
2(vi)	Write the formula for calculating ΔH from bond energy data. Using this formula, state what happens when: (a) sum of BDE of reactants > sum of BFE of products (b) sum of BDE of reactants < sum of BFE of products.	3	OR	Bond energy of H–H is 436 kJ/mol, I–I is 151 kJ/mol, and H–I is 299 kJ/mol. Calculate ΔH for: $\text{H}_{2(g)} + \text{I}_{2(g)} \rightarrow 2\text{HI}_{(g)}$. Show all working.	3
2(vii)	Define activation energy. State what collision theory says about effective and ineffective collisions.	3	OR	What two conditions must be met for a molecular collision to be effective? Why does a higher activation energy result in a slower reaction rate?	3
2(viii)	Define catalyst and catalysis. How does a catalyst increase the rate of a reaction without changing the overall enthalpy?	3	OR	What is an enzyme? How do enzymes in living organisms differ from industrial chemical catalysts in terms of specificity?	3
2(ix)	Define respiration. Why is respiration classified as an exothermic process?	3	OR	Write the word equation for aerobic respiration. Name all the reactants and products and state where the energy comes from.	3
2(x)	Differentiate between aerobic and anaerobic respiration. Write the word equation for each.	3	OR	Which type of respiration releases more energy? Why does the body use lipids as reserve energy stores when intense exercise is performed?	3
2(xi)	On a labeled reaction pathway diagram for an <i>exothermic</i> reaction, identify: (a) reactants level (b) products level (c) activation energy (d) enthalpy change.	3	OR	On a labeled reaction pathway diagram for an <i>endothermic</i> reaction, identify: (a) reactants level (b) products level (c) activation energy (d) enthalpy change. State the sign of ΔH .	3

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) Define exothermic and endothermic reactions. Explain the sign convention for ΔH in each case. [2] (b) Write thermochemical equations (with ΔH values) for any three reactions from Unit 8: one exothermic and two endothermic. Identify each type. [3]	5	OR	(a) Define thermochemical reaction and thermochemistry. Give the textbook example of a thermochemical equation. [2] (b) Classify each of the following as exothermic or endothermic and justify: (i) $\text{C}_{(s)} + \frac{1}{2}\text{O}_{2(g)} \rightarrow \text{CO}_{(g)}$, $\Delta H^\circ = -110.5 \text{ kJ}$ (ii) $\text{N}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{NO}_{(g)}$, $\Delta H^\circ = +180.5 \text{ kJ}$ (iii) freezing of water (iv) sublimation of dry ice (v) combustion of methane. [3]	5
3(ii)	(a) State the formula: $\Delta H^\circ = \text{Sum of BDE of reactants} - \text{Sum of BFE of products}$. Explain each term. [2] (b) Using bond energy data (H–H: 436 kJ/mol, Cl–Cl: 243 kJ/mol, H–Cl: 432 kJ/mol), calculate ΔH for the reaction: $\text{H}_{2(g)} + \text{Cl}_{2(g)} \rightarrow 2\text{HCl}_{(g)}$. Show all steps and state whether the reaction is exothermic or endothermic. [3]	5	OR	(a) Define bond dissociation energy and bond formation energy. State which process absorbs and which releases energy. [2] (b) Using bond energy data (H–H: 436 kJ/mol, F–F: 155 kJ/mol, H–F: 567 kJ/mol), calculate ΔH for the reaction: $\text{H}_{2(g)} + \text{F}_{2(g)} \rightarrow 2\text{HF}_{(g)}$. Show all steps and state whether the reaction is exothermic or endothermic. [3]	5
3(iii)	(a) Define activation energy. With reference to collision theory, explain why some collisions are effective and others are ineffective. [3] (b) Draw and label a reaction pathway diagram for an exothermic reaction showing: reactants, products, transition state, activation energy (uncatalyzed), activation energy (catalyzed), and enthalpy change. [2]	5	OR	(a) Define catalyst. Explain how a catalyst increases reaction rate without changing the overall enthalpy of the reaction. [3] (b) Draw and label a reaction pathway diagram for an endothermic reaction showing: reactants, products, transition state, activation energy, and enthalpy change. State the sign of ΔH and explain why products are at a higher energy level than reactants. [2]	5
3(iv)	(a) Define respiration and state why it is an exothermic process. Distinguish between aerobic and anaerobic respiration with word equations. [3] (b) Explain the role of lipids as reserve energy stores. Under what conditions does the body switch from oxidizing glucose to oxidizing lipids? [2]	5	OR	(a) Compare aerobic and anaerobic respiration under four headings: oxygen requirement, reactants, products, and energy released. [3] (b) Explain how enzymes act as biological catalysts. How does this connect to the concept of activation energy? [2]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Chapter 8 — Energetics (pp. 127–133)

End of Question Paper